

Software Engineer with strong experience in C++ and performance-critical software. Proven track record delivering production features on large-scale software platforms. Skilled in debugging, system architecture, automation, and cross-team technical leadership. Proven flexibility, capability, and passion for working with new technologies and programming languages, quickly adapting to different environments and contributing effectively across diverse software domains.

WORK EXPERIENCE

•Motorola Solutions

March 2025-Present

Senior Software Engineer

Cork, Ireland

- Developing and maintaining C++ software running on radio communication devices
- Contributing to internal test automation tools written in C# with Python scripting
- Debugging and fixing C++ software code
- Working closely with embedded engineers, QA, and product managers to deliver high-quality features in AGILE environment

•Forcepoint

July 2024-February 2025

Software Engineer III

Cork, Ireland

- Designed and implemented major components of a Manifest V3 browser extension (JavaScript) used for user interaction and security workflows
- Automated operational workflows using PowerShell, reducing manual workload and improving reliability
- Collaborated with cross-functional teams, including software engineers, QA engineers, and product managers, following AGILE methodologies to ensure high-quality deliverables and adherence to project timelines.

•Qualcomm

Oct 2022-June 2024

Software Engineer

Cork, Ireland

- Developed production-quality C++ software for embedded systems and internal tools
- Wrote and maintained unit tests using GoogleTest
- Created UML diagrams, technical documentation, and internal design specifications
- Led technical implementation of multiple projects, trained colleagues and ran technical interviews

•ResilTech S.R.L.

Oct 2021-July 2022

Software Engineer

Pisa, Italy

- Designed and developed C and C++ software for safety-critical systems
- Development and maintained of Python, Bash, Makefile and Perl scripts for automation and testing
- Designed testbenches for safety platforms using Verilog/SystemVerilog

•SECO S.P.A.

Oct 2017-June 2018

Quality assurance engineer

Arezzo, Italy

- Analysed and validated C++ codebases
- Automated testing using Bash scripts
- Performed mathematical and statistical analysis of gaming algorithms

RESEARCH

•University of Pisa

Sept 2020-July 2021

Master's degree thesis: Implementation of a CNN on a soft GPU through OpenCL

Pisa, Italy

- Goal of the thesis was to implement a Convolutional Neural Network on a soft GPU using OpenCL and C++ languages.
- Research on GPU software execution for Single Instruction Multiple Data execution.
- Development of OpenCL scripts used to execute CNN layers.
- Development of a C++ code used to handle the GPU execution, like indicate which OpenCL script execute.
- The system was implemented on a development board containing an FPGA on which the GPU was implemented, and a CPU on which a C++ code was running.

•University of Siena

June 2017-Oct 2017

Bachelor's degree thesis: Integration of heterogeneous sensors on a self-driven drone

Siena, Italy

- Goal of the thesis was to develop a self-driven boat used to acquire data from a water environment.
- Development of the code used to acquire data from the sensors.
- Development of the code used to send the data on land using LoRa protocol.
- The system was implemented on a Arduino board.

EDUCATION

•**Master Degree in Electronic Engineering**

2018-2021

University of Pisa, Italy

•**Bachelor Degree in Computer Engineering**

2014-2017

University of Siena, Italy

TECHNICAL SKILLS AND INTERESTS

Programming Languages: C++, C, Python, C#, Bash

Software tools: git, GitHub, Linux, CMake, Makefile, UML, Valgrind, GNU debugger (gdb), lcov, Boost library

Informatic Knowledge: Object Oriented Programming, Computer Architectures, Operating Systems, Informatics fundamentals, Data structures, Algorithms, Software Engineering

Basics: Mathematical Analysis, Algebra, Probability Theory, Physics, Electronics

Languages: Italian, English